CHAPTER 1 · GRADE 4 · WEEK 1

Energy Causes Change

Discovery / Evidence

Where can we observe evidence of energy causing change?

ENERGY → CHANGE → ?

Unit 1 · Energy Transfer & Transformation



TURN & TALK BQ

Big Question

Where can we observe evidence of energy causing change?

Keep it in mind all chapter. We will come back to it.

Think: What is ONE change you have seen happen?

WRITE N

BQ

Big Question — My first thinking

Guided Notes p. 1

Where can we observe evidence of energy causing change?

My first thinking

Write 1–2 quick ideas. No wrong answer yet.

OBSERVE D

What changed?

Reader p. 1 — you walk into a dark room and flip a switch.

DARK room



hard to see

BRIGHT room



light fills the room

Something became different. That is a change.

TURN & TALK D

I notice… I wonder…

I NOTICE

The room is different now.
Light is everywhere.

I WONDER

What made the light change so fast?

Share one notice and one wonder with a partner.

OBSERVE D

Electric Pencil Sharpener

Observe the pencil BEFORE we sharpen it.

BEFORE

A dull pencil — tip is flat, and it is quiet.

Look closely. What do you see? What do you hear?

PREDICT D

Predict: what will change?

Think before we turn it on.

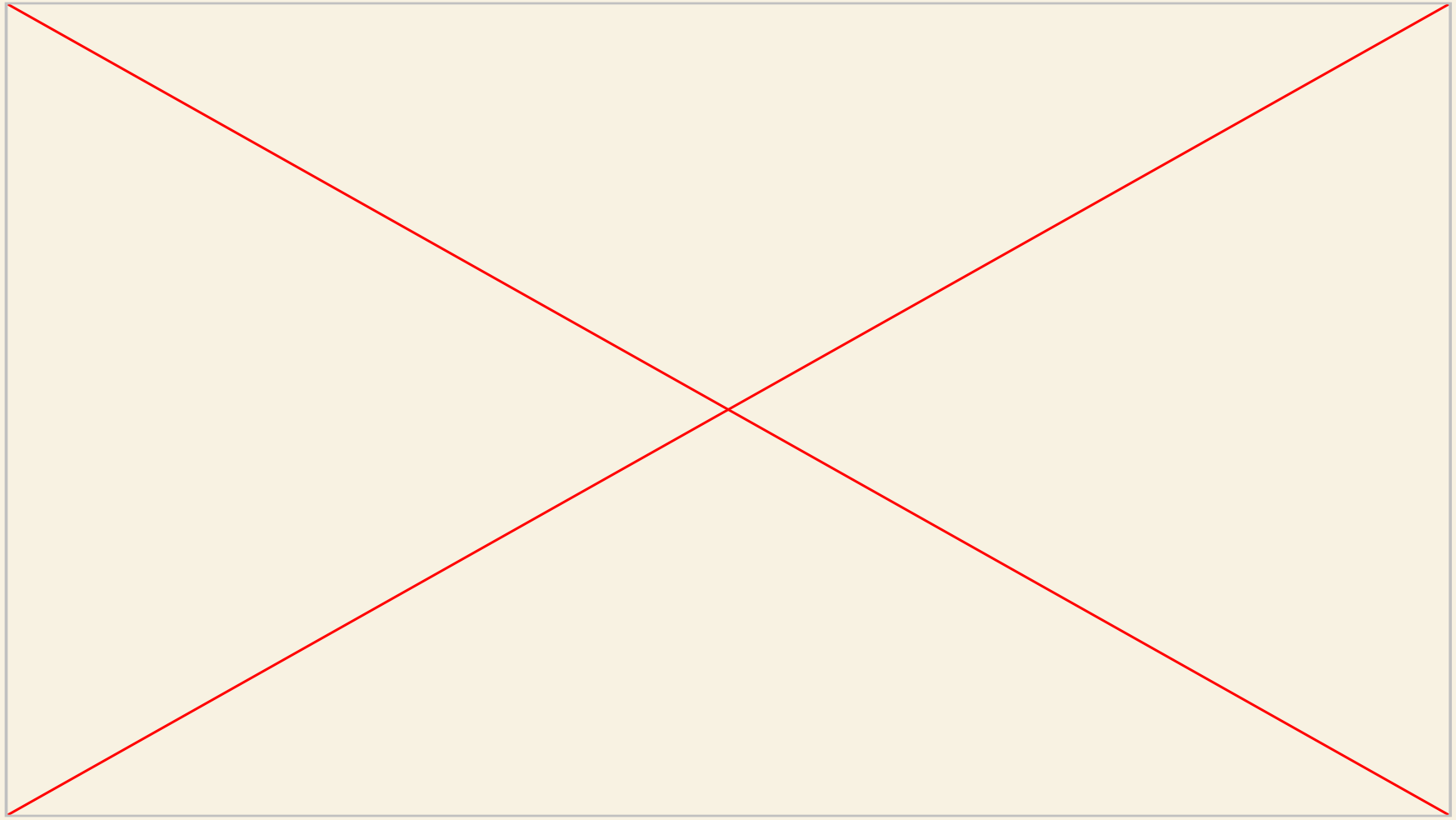
When the sharpener runs, the pencil will…

get _____

Make a prediction. Say it to your partner.

WATCH & OBSERVE — Pencil Sharpening

Find out: What changes about the pencil? What do you see and hear?



OBSERVE D

What changed?

BEFORE	AFTER	CHANGES
dull tip · quiet	shorter · sharper · humming	• the tip is sharper • the pencil is shorter • it makes a sound

EXPLAIN D

How do we know energy was involved?

Energy can be invisible. We look for the change.



The pencil got shorter and sharper and made a sound.
Those changes are clues that energy was involved.

V

Vocabulary — Energy & Change

Reader p. 1

VOCABULARY

TEACH

energy

the ability to cause change



ENERGY → CHANGE

Reader p. 1

VOCABULARY

TEACH

change

to become different

BEFORE → AFTER

Reader p. 1

WRITE N

V1

Vocabulary — Energy & Change

Guided Notes · Reader p. 1

energy — the ability to cause _____ .

change — to become _____ .

Write the missing word in each blank.

CHECK N

V1

Vocabulary — Energy & Change

Guided Notes · Reader p. 1

energy — the ability to cause change .

change — to become different .

change · different

RECITE R

What is energy?

Reader p. 1

> Energy is the ability to cause change.

Say it together: “Energy is the ability to cause change.”

Look at the seashore

Reader p. 2 — the seashore changes all around us.



Read Reader p. 2.
Waves move. Sand shifts.
Water makes sound.
What are these changes a clue about?

ENERGY → CHANGE → ?

WRITE N

N1

Energy Causes Change

Guided Notes · Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing _____ of energy.

Write the missing word in the sentence. Leave the last box of your model empty for now.

N1 Energy Causes Change

Guided Notes · Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing evidence of energy.

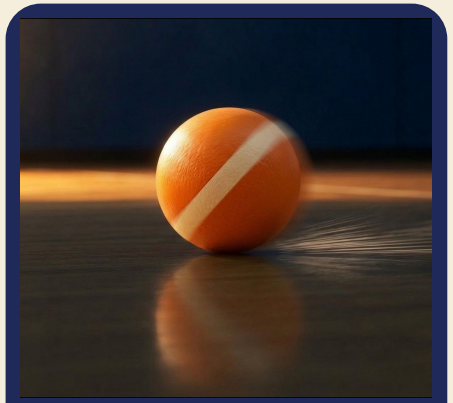


Now write the same word in the last box of your model.

R

Energy has different forms

Reader p. 3 — think of a spinning carousel.



motion



sound



light



heat



electrical energy

On the carousel, the horses spin and move up and down.
That is motion.
Find the other forms in the Reader.

OBSERVE C

Clap three times

What forms of energy do you observe?

When you clap…

What do you SEE?
What do you HEAR?
What do you FEEL in your hands?

WRITE N

N2 Energy Occurs in Different Forms

Guided Notes · Reader p. 3

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	
sound	
light	
heat	
electrical energy	

V

Vocabulary — Stored Energy

Reader p. 4

VOCABULARY

TEACH

stored energy

energy that has the ability to cause change at a later time



Reader p. 4

WRITE N

V2

Vocabulary — Stored Energy

Guided Notes · Reader p. 4

stored energy — energy that has the ability to cause change at a _____ time.

Write the missing word in the blank.

CHECK N

V2

Vocabulary — Stored Energy

Guided Notes · Reader p. 4

stored energy — energy that has the ability to cause change at a later time.

later

PREDICT R

Does a still battery have energy?

Reader p. 4 — think about a battery inside a flashlight.



A battery sits still in a flashlight that is switched off.

Yes or no? Make a prediction.
Tell your partner why.

R

The stored energy of position

Reader p. 5 — an object held up can cause a change later.



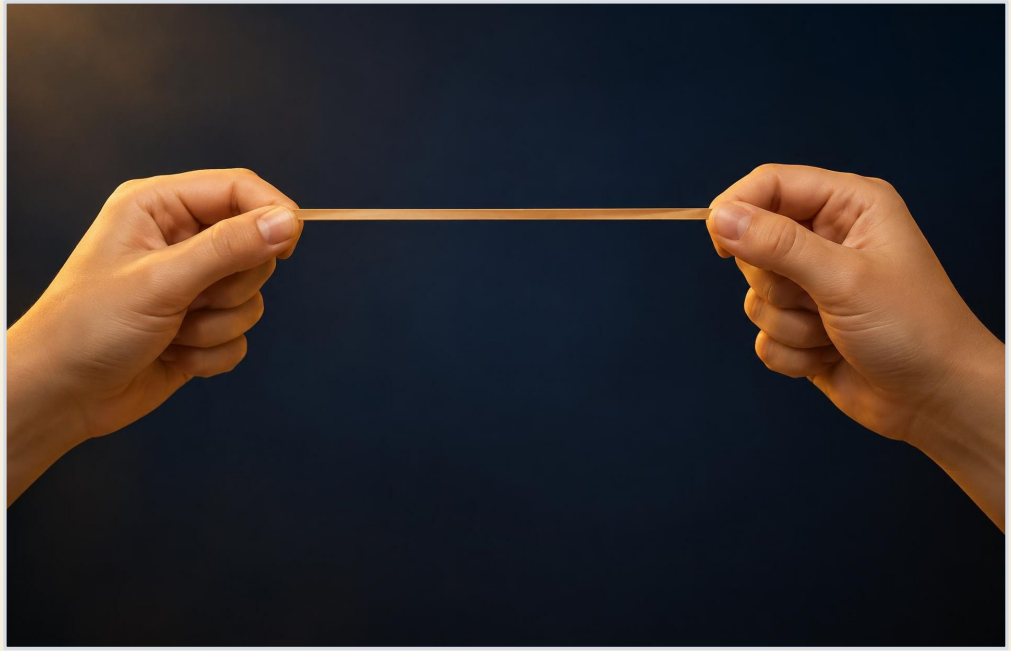
Raised apple → let go → ?

An apple held above the ground has stored energy
because of its position. Imagine you let go. What will happen?

Stored energy can cause a change later

Reader p. 4 — back to our battery: the chemicals inside it store energy.

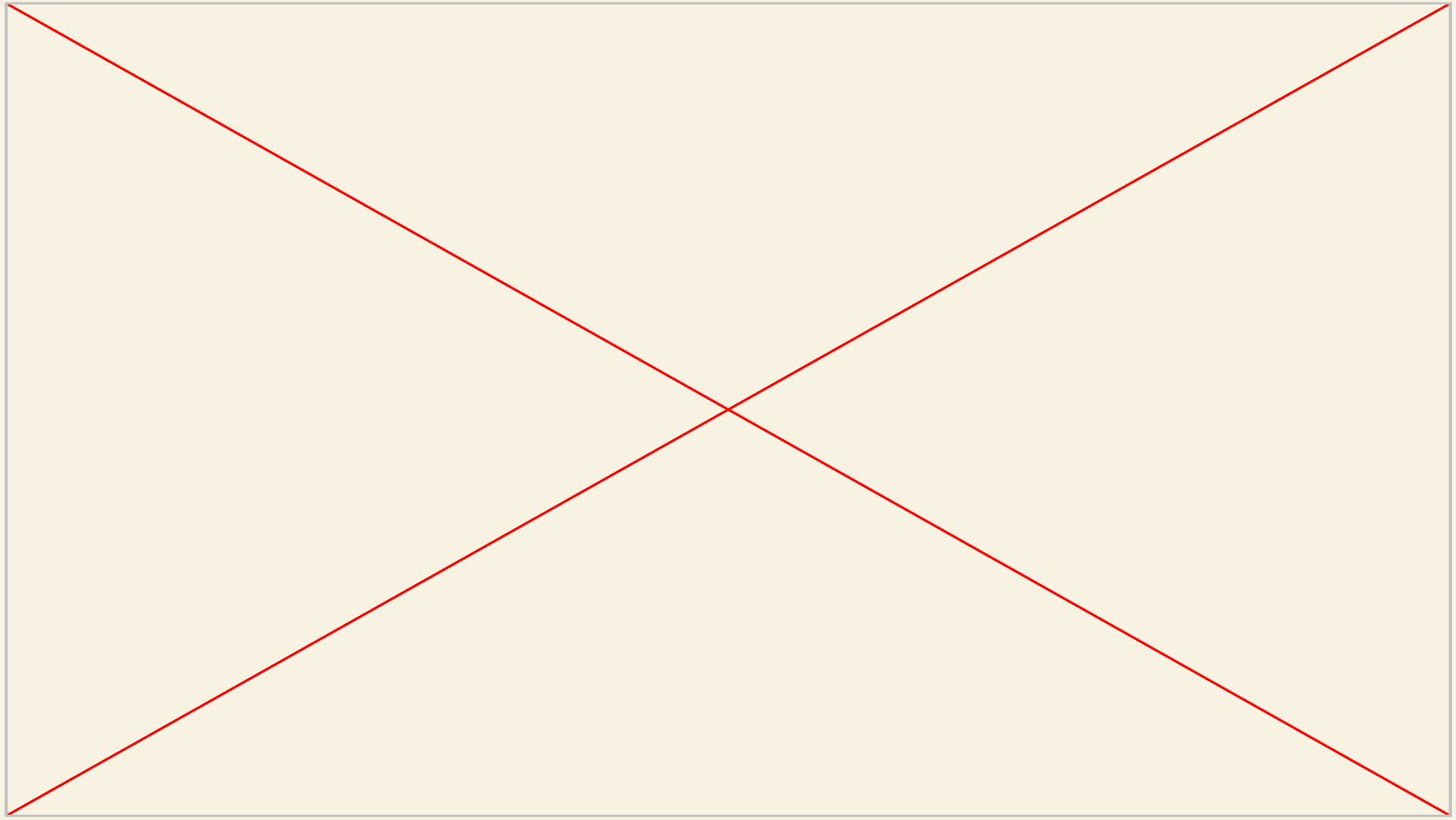
STORED ENERGY → RELEASE → CHANGE



Stretch a rubber band — it now stores energy.
Let it go. What change will you see and hear?

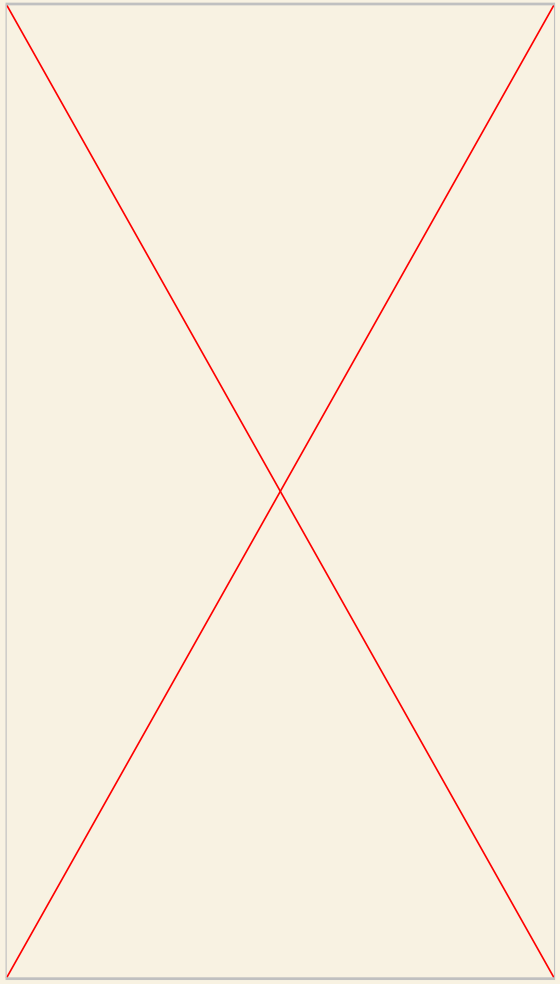
WATCH & THINK — Stored Energy

Find out: How can something that is still cause a change later?



WATCH & THINK — Still Objects Can Have Energy

Follow along: How can something that is not moving still cause a change later?



WRITE N

N3 Objects Can Store Energy

Guided Notes · Reader pp. 4–5

STORED ENERGY → RELEASE → CHANGE

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object (apple)	

Objects also have stored energy because of their _____ .

CHECK N

N3 Objects Can Store Energy

Guided Notes · Reader pp. 4–5

STORED ENERGY → RELEASE → CHANGE

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	*the flashlight lights up*
stretched rubber band	*it snaps and makes a sound*
raised object (apple)	*it falls down*

Objects also have stored energy because of their position .

position · Sample answers shown - yours may differ.

EVIDENCE R

Recognizing evidence

Reader pp. 1–5 — think about the clues we can see, hear, or feel.

Energy can be invisible. What clues could show that energy is at work?





Think of a change you have seen. What did you see, hear, or feel?

N4

Recognizing Evidence

Guided Notes · Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER
draw	draw

What changed?

WRITE N

N4

Check Your Thinking

Guided Notes · Reader pp. 1–5

CHECK YOUR THINKING

How do we know energy is involved if we cannot see energy itself?

Start with: "We know because we see …"

CHECK C

We know because we see a change

SAMPLE

“We know because we see a change — like a ball starting to move or a light turning on.”

Any change you observe is evidence of energy.

EVIDENCE C

Which change shows energy at work?

A toy car starts to roll.

A rock keeps sitting still.

A room stays dark.

Pick one. Tell your partner why.

WRITE N

BQ

BQ · Now I Think

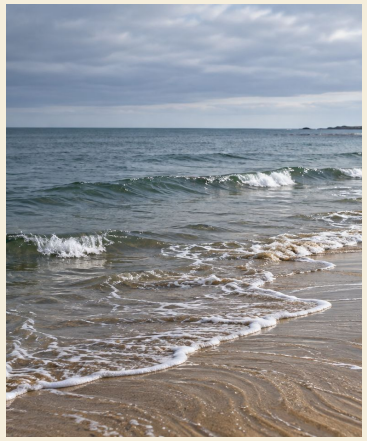
Guided Notes p. 3

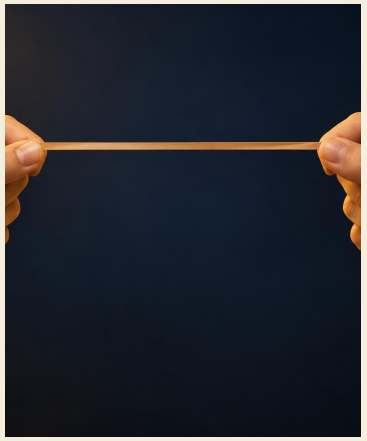
BIG QUESTION

Where can we observe evidence of energy causing change?









Now I think…

C

Review 1 — concept

Energy is best described as —

A a visible glow

B something only batteries have

C the ability to cause change

D a kind of fuel

CHECK C

Review 1 — answer

Energy is best described as —

A a visible glow

B something only batteries have

C the ability to cause change

D a kind of fuel

Energy is the ability to cause change (Reader p. 1).

C

Review 2 — visual

A pencil is dull, then you sharpen it. What change do you observe?

A It gets shorter and sharper

B It gets longer

C It changes color

D It disappears

CHECK C

Review 2 — answer

A pencil is dull, then you sharpen it. What change do you observe?

A It gets shorter and sharper

B It gets longer

C It changes color

D It disappears

The pencil becomes shorter and sharper — a change.

C

Review 3 — scenario

Which object has stored energy that can cause a change LATER?

A A ball already rolling

B A toy already moving

C A flashlight already on

D A stretched rubber band

CHECK C

Review 3 — answer

Which object has stored energy that can cause a change LATER?

A A ball already rolling

B A toy already moving

C A flashlight already on

D A stretched rubber band

The rubber band stores energy; it changes when released.

C

Review 4 — comparison

A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

A only light

B motion and sound

C motion and silence

D sound and stillness

CHECK C

Review 4 — answer

A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

A only light

B motion and sound

C motion and silence

D sound and stillness

The spinning is motion; the music is sound (Reader p. 3).

C

Review 5 — concept

How do we know energy is involved if we cannot see it?

A We observe a change happen

B Energy can always be seen

C We guess without evidence

D Nothing can change

CHECK C

Review 5 — answer

How do we know energy is involved if we cannot see it?

A We observe a change happen

B Energy can always be seen

C We guess without evidence

D Nothing can change

We see a change — like movement — as evidence.

C

Review 6 — scenario

A battery sits still in a flashlight. Does it have energy?

A No — still objects have no energy

B Only when it is moving

C Yes — stored energy that can cause change later

D Only when the flashlight is on

CHECK C

Review 6 — answer

A battery sits still in a flashlight. Does it have energy?

A No — still objects have no energy

B Only when it is moving

C Yes — stored energy that can cause change later

D Only when the flashlight is on

A battery stores energy. It can cause a change later.